Chapter Two

Data Communication Layer

Compiled by Yilkal B.

Chapter Outline

- *Architectures*
- *Protocol standards*
- *OSI 7-layer reference model and concepts of open standards*
- *Motivations for standards*
- *ISO reference model*
- *Open standard*
- *Internetworking with TCP/IP Structure*

Communication Architecture

- ***Network architecture*** *is a* ***design of communication networks*** *for specification of a network's physical components and their functional organization and configuration, its operation principles and procedures as well as data formats use.*
- ***Network architecture*** *is predominantly expressed by its use of the internet protocol suits.*

Data Communication Architecture:

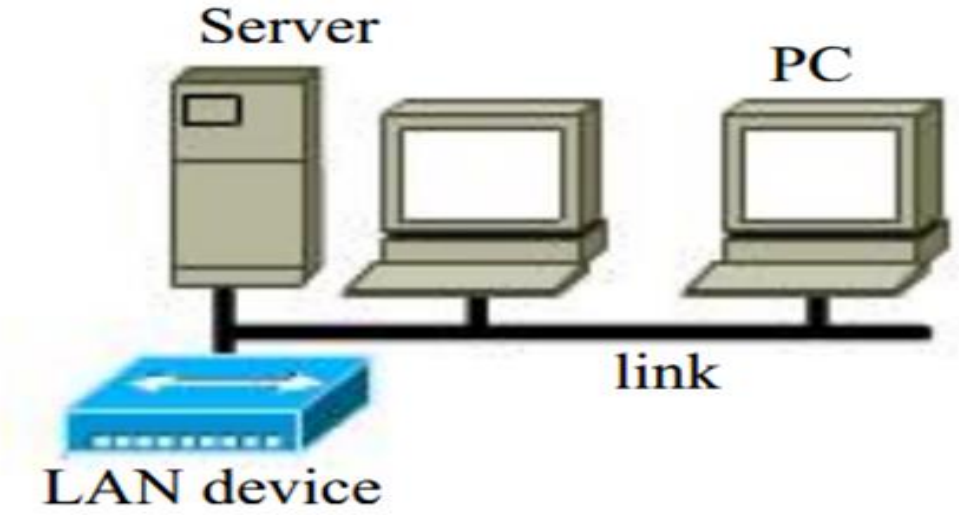
Network Line Configuration: *defines the attachment of communication devices to a link*

- *Or both* ***point to point*** *and* ***multipoint*** *connection are line configuration to connect two or more devices in link.*
- ***Link :*** *a physical communication pathway that transfers data from one device to another.*
- ***point-to-point:-*** *a line configuration where two and only two devices are connected by a dedicated link.*
- *It is* ***unicast connection*** *because there is dedicated link between the two devices*

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Multipoint: *– a line configuration in which more than two specific devices share a single link.*

- *In multipoint connection a single link can be shared between multiple devices.*
- *Single channel shared*
- *Single transmitter but multiple receiver*
- *No dedicated line*



Network Design Models

There are basically have 2 network design models

A. Flat model:- it is single collision domain and not hierarchical divided.

It has some advantage:

- Easy to design
- Easy to implement
- It good for small network
- East to maintain and diagnosis

It has some disadvantage:

- Poor security
- No redundancy
- Less speed
- No scalability
- No networks layers

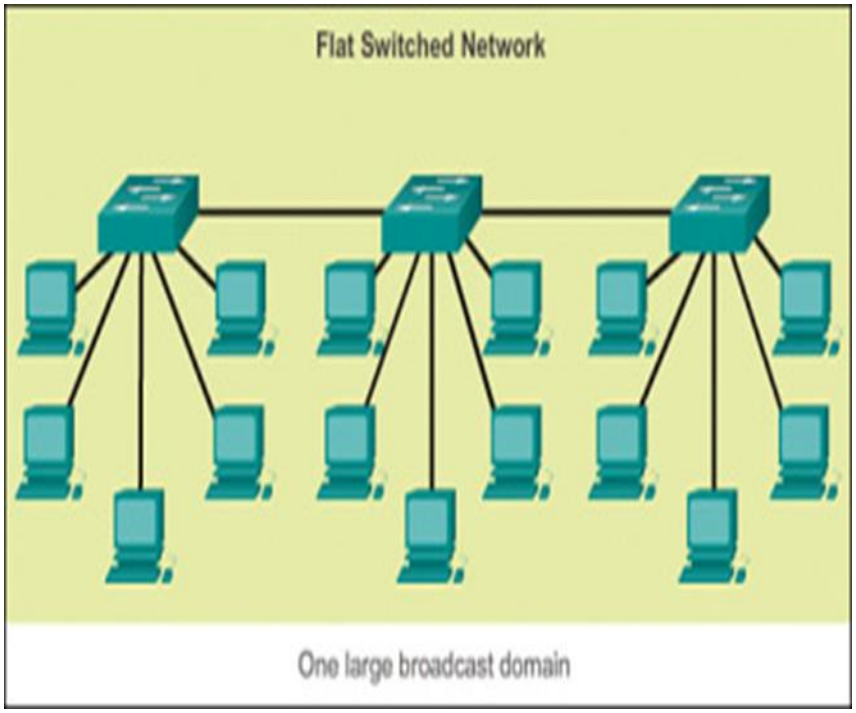


fig.1 Flat switch network

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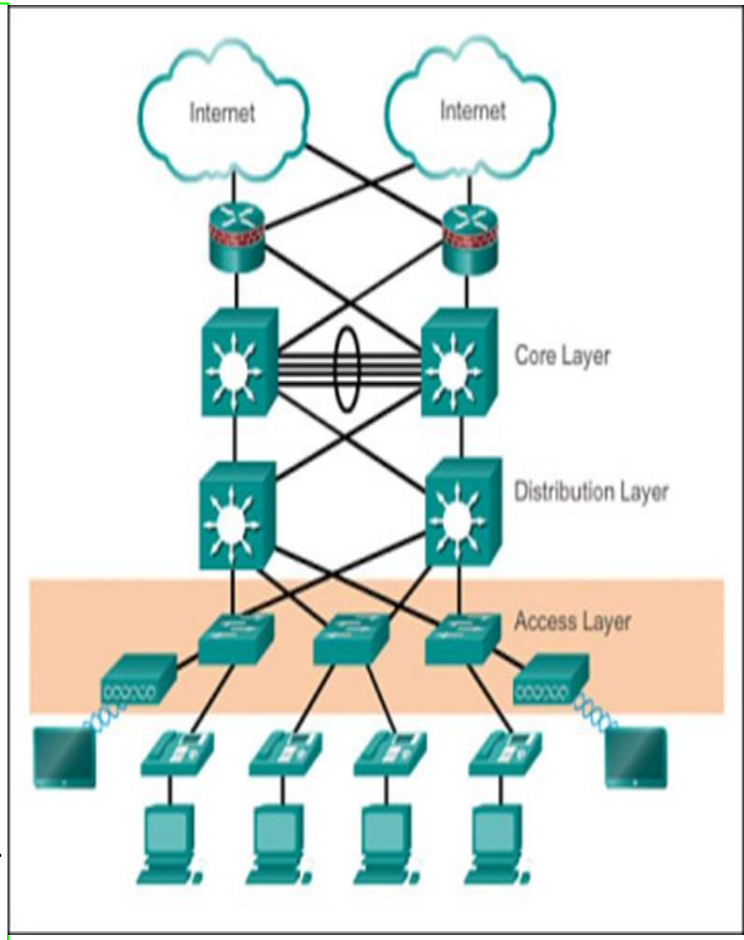
B, Hierarchical Model:– *it is a network model that has* ***three*** *network layers or divisions and each layers has their own specific functions.*

The three layers are:

I. ***Access layer:*** *used* ***for service availably port security*** *on layer 2* ***switch*** **.**

II. ***Distribution layer:*** *aggregate data received from access layer and switches from sending to* ***core layer,*** *redundancy and load balancing, control broadcast domain*

III. ***Core layer control:*** *it is a* ***network backbone,*** *it is fast speed layer it connect multiple campus network and data center*



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- ***It has some advantage of hierarchical model:***
  - *more security*
  - *Better speed or performances*
  - *Facilitate scalable performance of network*
  - *Good for big network*
  - *Easy to troubleshoot*
- ***It has some disadvantage of hierarchical model:***
  - *Difficult to design*
  - *Difficult to implement*

Network Protocols and Standards

- ***A protocol*** *is a set of rules that governs data communications*
- *a protocol defines* ***what*** *is communicated,* ***how*** *it is communicated, and* ***when*** *it is communicated.*
- *for instance, for one computer to send a message to another computer, the* ***first computer*** *must perform the following general steps*
  - *break the data into small sections called packets*
  - *add addressing information to the packets identifying the destination computer*
  - *deliver the data to the network card for transmission over the network*
- *The* ***receiving computer*** *must perform the same steps, but in reverse order*
  - *accept the data from the NIC*
  - *remove transmitting information that was added by the transmitting computer*
  - *reassemble the packets of data in to the original message*

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- ***the key elements of a protocol are:***
  - ***syntax:*** *refers to the structure or format of the data*
  - ***semantics:*** *refers to the meaning of each section of bits*
  - ***timing:*** *refers to when data should be sent and how fast they can be sent*

- ***Some important characteristics of a protocol are:***

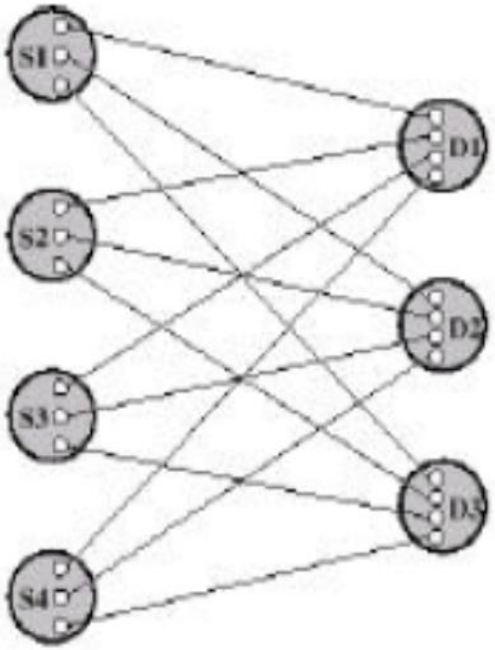
- ***Direct /indirect:***
  - *If two systems share a point-to-point link - **direct***
  - *systems connect through a switched communication network – **indirect***

- ***Monolithic/structured***
  - *Communications is a complex task for a single unit*
  - *One big protocol vs. many structured protocols function as one.*

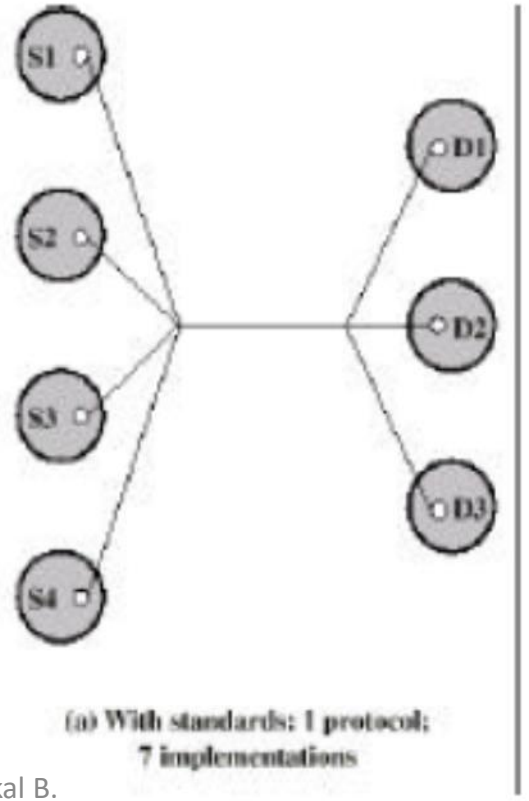
- ***Symmetric/asymmetric:*** *Communication between peer entities Vs Client/Server*

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- ***Standard /nonstandard***
  - *Non-standard protocols built for specific computers and tasks*



(a) Without standards: 12 different protocols; 24 protocol implementations



(a) With standards: 1 protocol; 7 implementations

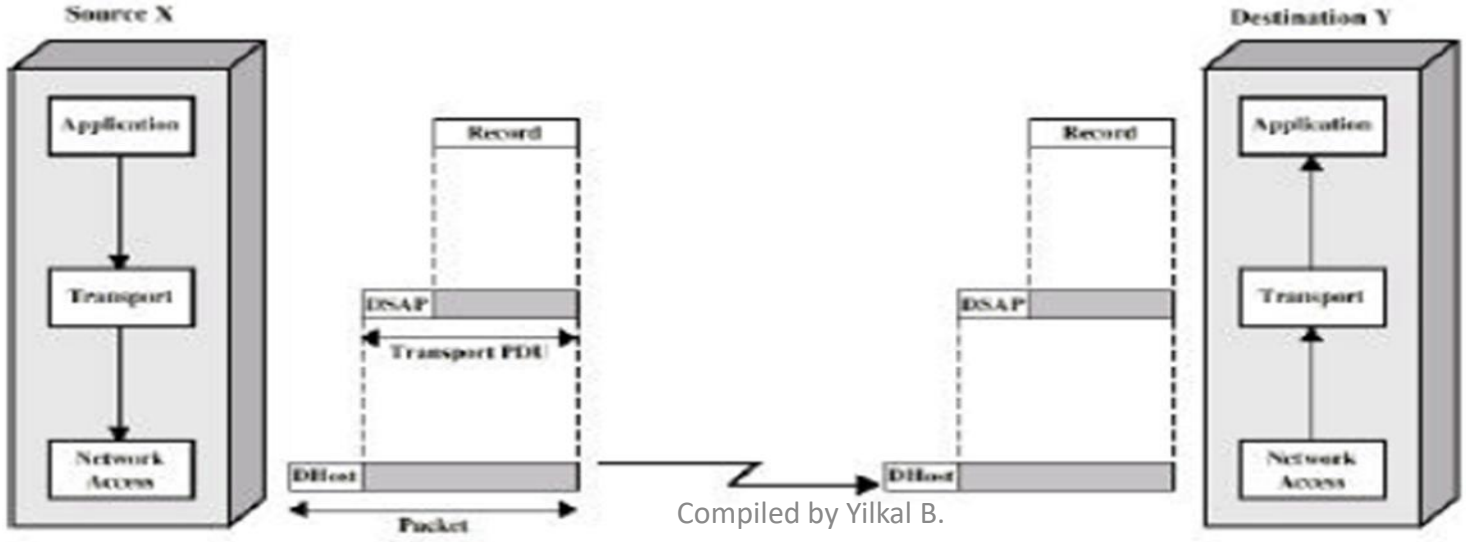
Functions of Protocols

- *Each device must perform **the same steps the same** way so that the data will arrive and **reassemble** properly; if one device uses **a protocol with different** steps, the two devices will **not be able** to communicate with each other*
- *The following are categories of functions that form the basis of protocols*
  - *encapsulation*
  - *segmentation and reassembly*
  - *connection control*
  - *addressing*
  - *multiplexing*
  - *transmission services*

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1. Encapsulation

- the *inclusion or adding of control information* to data is called **encapsulation.**
- Each frame contains not only data but also **control information**
- such **control information** falls into **three categories: address** (sender/receiver), **error-detecting code** and **protocol control** (information about protocol functions)



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2. Segmentation

- by the sender also called **fragmentation** and **reassembly** by the receiver a block of data for transmission is of some bounded size
- at the **application level**, we refer to a logical unit of data transfer as a message
- lower level protocols may need to break the data up into blocks of some bounded size this process is called **segmentation**
- **reasons for segmentation:**
  - **the communication network may only accept** blocks of data up to a **certain size**
  - **error control** may be more efficient with **a smaller frame size**; fewer bits need to be retransmitted when a frame gets corrupted
  - **facilitates more equitable access** to **shared transmission facilities**

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- ***disadvantages***
  - *since a frame contains* ***certain amount of control information****, the smaller the block size, the greater the percentage* ***overhead***
  - ***frame arrivals may generate*** *an interrupt that must be serviced; hence smaller blocks result* ***in more interrupts***
  - ***more time is spent processing*** *smaller, more numerous frames*
- *the counterpart of segmentation is* ***reassembly***
- *on receipt, the receiving device must reassemble the segmented data into a* ***message*** *appropriate to the application*

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3. Connection control : *two types of protocols (services)*

- ***connectionless service***
  - *the packets are sent from one party to another with no need for connection establishment or connection release*
  - *the packets **are not numbered**, they may be delayed, lost, or arrive out of sequence; there is no acknowledgement either.*
  - ***UDP** (User Datagram Protocol), one of the transport layer protocols, is connectionless*
  - ***good for one time transaction**; e.g., email*

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- ***connection-oriented service:*** *in connection-oriented data transfer, a logical association or* ***connection is established between*** *the communicating computers (devices).*
  - *long sustained session*
  - *orderly and timely delivery of packets, e.g., FTP*

three phases are involved

- ***connection establishment*** *(agreement to exchange data)*
- ***data transfer*** *(data and control information exchanged)*
- ***connection termination*** *(termination request) - by any of the two parties*

- ***TCP*** *(Transport Control Protocol) is connection-oriented*
- *the key characteristics of connection-oriented data transfer is that* ***sequencing*** *is used each side sequentially numbers the frames that it sends to the other side.*

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- because each side remembers that it is engaged in a **logical connection**, it can keep track of both outgoing numbers, which it generates, and incoming numbers which are generated by the other side

Sequencing supports three main functions

- **ordered delivery:** frames may not arrive in the order in which they were sent, because they may traverse different paths
- **flow control:** a receiving station has to limit the amount or rate of data that is sent by a transmitting station
  - the simplest form of flow control is **stop-and-wait procedure** in which each frame must be acknowledged before the next can be sent
  - more efficient protocols involve some form of credit provided to the transmitter, which is the amount of frames that can be transmitted **without an acknowledgement**; e.g. sliding window technique

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❖ ***error control:*** *error control is implemented as two separate functions; error detection and retransmission*

- *if an error is detected, the receiver discards the frame*
- *upon failing to receive an* ***acknowledgement*** *to the frame in a specified reasonable time, the sender retransmits the frame*
- *some protocols also employ error correction which enables the receiver not only to detect errors but, in some cases,* ***to correct them***

4. Addressing

- ***a unique address*** *is associated with each end-system in a configuration (e.g. workstation, server) and each intermediate system (e.g. router)*
- *an example is an IP address in TCP/IP connection*

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5. multiplexing: *occurs when* ***multiple*** *connections share a* ***single connection*** *(multiple access).*

6. transmission services

- *a variety of additional services can be provided*
- ***priority:*** *messages such as control messages may need to get through to the destination station with minimum delay*
- ***security:*** *security mechanisms, restricting access, may be invoked*

❖ ***Protocols in a layered architecture :*** *protocols that work together to provide a layer or layers of the model are known as* ***a protocol stack or suite****, e.g. TCP/IP*

- *each layer handles* ***a different part of the communications process*** *and* ***has its own protocol***

The principal motivation for the development of the OSI model

- To provide a framework for standardization.
- Within the model, one or more protocol standards can be developed at each layer.
- **Changes in** standards in one layer **need not affect** already existing software in another layer.
- The **design principle of information hiding** is used:
  - **Lower layers** are concerned with **greater levels of detail;**
  - **upper layers** are independent of these details.
- To dominate computer communications, eventually replacing proprietary protocol implementations and rival multivendor models such as TCP/IP.
- This is **no happened**. Instead, the **TCP/IP architecture** has come to dominate.

Models

- *For two computers, in a computer network, to communicate must talk the same language.*
  - *follow the same protocol*
- ***A network protocol** is a formal set of rules, conventions and data structure that governs how computers and other network devices exchange information over a network.*
- ***In earlier days**, many of the networks that were built used different hardware and software implementations*
  - *they were **incompatible** and it became **difficult for networks** using different specifications to communicate with each other*

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- To address the problem of networks being **incompatible** and **unable** to communicate with each other, we **need some way of model** .

- A widely accepted structuring technique is **layering**
- The communications functions are **partitioned into a hierarchical** set of layers.
- Each layer performs a related subset of the functions
- **It provides services** to the **next-higher layer** and **accepts service for the lower layer**
- Each layer covers different **network activities**, equipment, or **protocols**
- This separation into smaller more manageable functions is known as **layering.**
- **Layering** specifies different functions and services as data moves from one computer through the **network cabling** to another computer

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- ***The layers are separated from each other by boundaries*** *called* ***interfaces***
- *All requests are passed from one layer,* ***through the interface****, to the next layer*

E.g. OSI and TCP/IP

Why layering?

- *reduces complexity (one big problem to smaller ones)*
- *standardizes interfaces (between layers)*
- *facilitates modular engineering (different teams work on different modules)*
- ***assures interoperable technology***
- *accelerates evolution of networking technology*
- *simplifies teaching and learning*

OSI (open system interconnection) Reference Model

- *It was developed by the **International Organization for Standardization** (ISO) in **1970** . /OSI=open system interconnection*
- ***It is a model f**or a computer protocol architecture and as a **framework for developing protocol standards.***
- *An ISO standard that covers all aspects of network communications is the Open Systems Interconnection (OSI) model.*
- *is now considered **the primary Architectural model for inter-computer communications.***
- ***is a descriptive network scheme.***
- *It ensures greater compatibility and interoperability between various types of network technologies.*
- ***Comprises of seven layers***
- *Defines **how each layer communicates** and works with the layers immediately above and below it*

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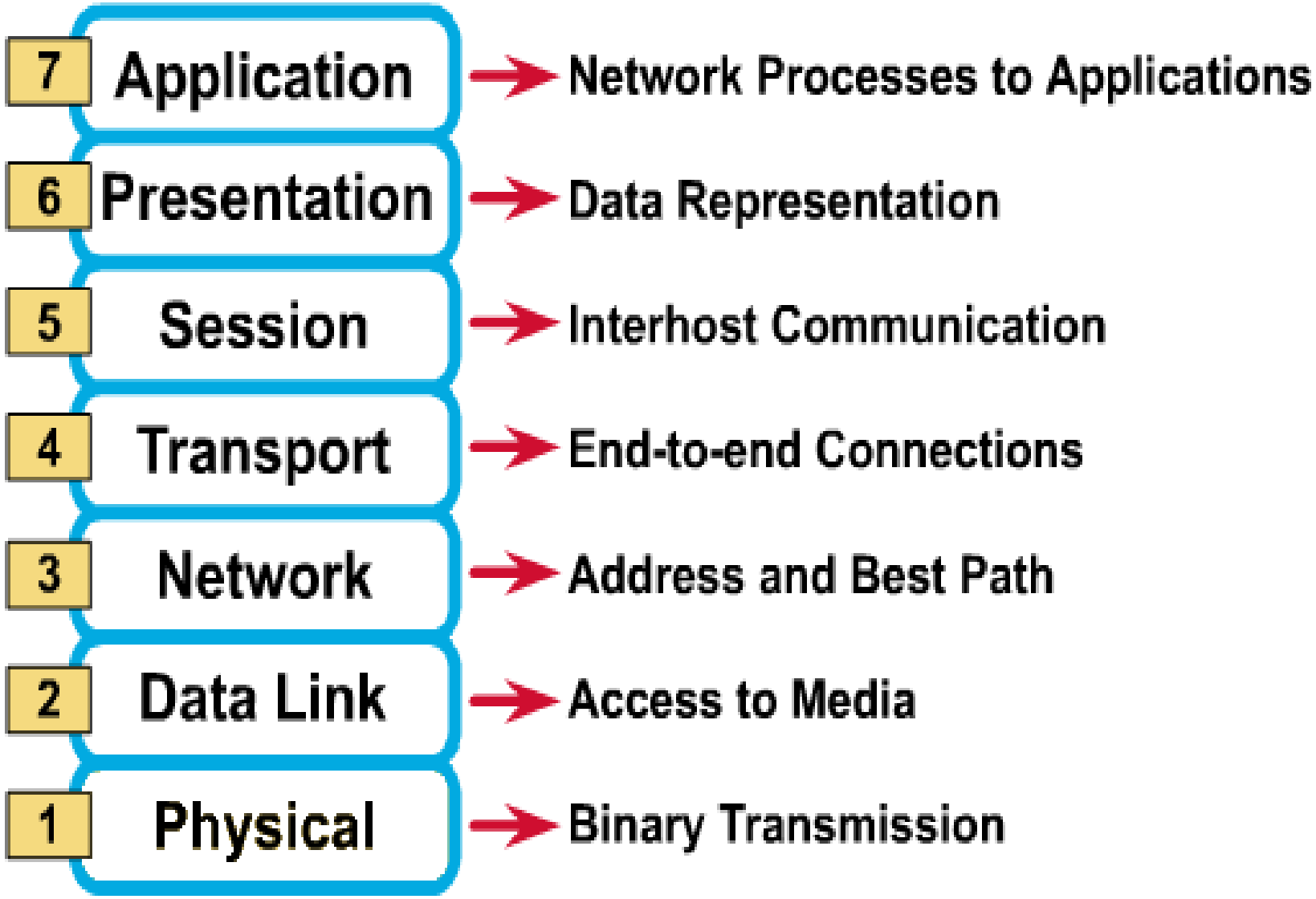
Why OSI important

- Network communication is broken into smaller, more **manageable parts**.
- Allows **different types** of network hardware and software to communicate with each other.
- **All layers are independent** and **changes does not affect other layers**.
- Easier to understand network communication.

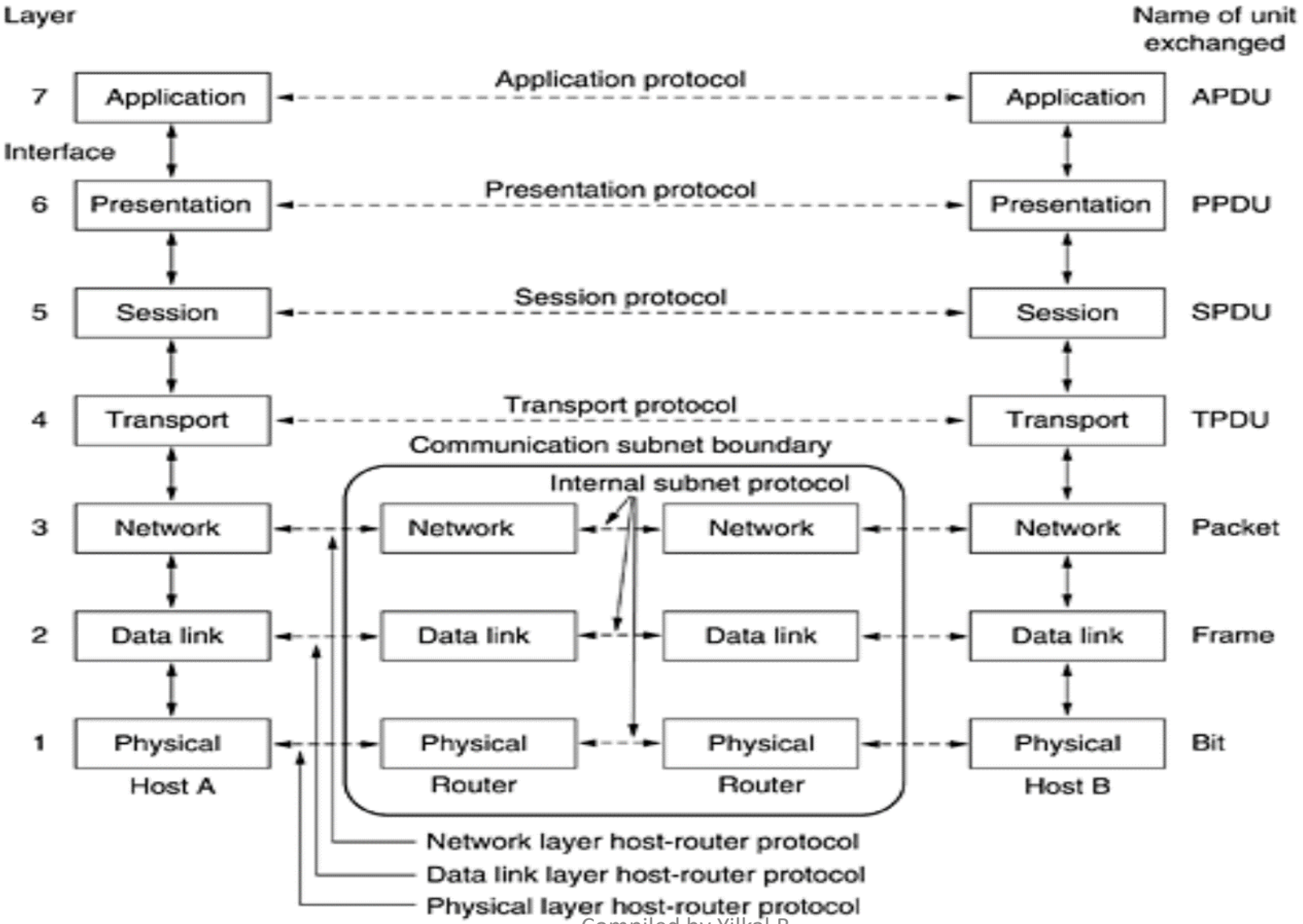
The Seven OSI Reference Model Layers

- Each layer provides a **service to the layer above** it in the protocol specification.
- Each layer communicates with the same layer's software or hardware on other computers.
- The **lower 4 layers** (**transport, network, data link** and **physical** : Layers 4, 3, 2, and 1) are concerned with the flow of data from end to end through the network.
- **The upper three layers** of the OSI model (**application, presentation** and **session**: Layers 7, 6 and 5) are **orientated more toward services to the applications**.
- Data is **Encapsulated** with the necessary protocol information as it moves down the layers before network transit.

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7 Application
Network Processes to Applications
6 Presentation
Data Representation
5 Session
Interhost Communication
4 Transport
End-to-end Connections
3 Network
Address and Best Path
2 Data Link
Access to Media
1 Physical
Binary Transmission

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Layer
Name of unit exchanged
7
Application
Application protocol
Application
APDU
Interface
6
Presentation
Presentation protocol
Presentation
PPDU
5
Session
Session protocol
Session
SPDU
4
Transport
Transport protocol
Transport
TPDU
Communication subnet boundary
Internal subnet protocol
3
Network
Network
Network
Network
Packet
2
Data link
Data link
Data link
Data link
Frame
1
Physical
Physical
Physical
Physical
Bit
Host A
Router
Router
Host B
Network layer host-router protocol
Data link layer host-router protocol
Physical layer host-router protocol

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The principles applied to arrive at the seven layers

- *Each layer should perform a **well-defined function**.*
- *A layer should be created where a different abstraction is needed.*
- *The function of each layer should be chosen with an eye toward defining internationally standardized protocols.*
- *The layer boundaries should be chosen to **minimize the information flow across** the interfaces.*
- *The number of layers should be **large enough** that **distinct functions need not be thrown together in the same layer out of necessity** and **small enough** that the architecture does not become **unwieldy**.*

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Layer 7: Application

- The application layer is the OSI layer that is **closest to the user.**
- It provides network services to the user's applications.
- contains all the **higher level protocols** that are commonly needed by users

Layer 6: Presentation

- The presentation layer ensures that the information that the application layer of one system sends out is **readable by the application** layer of another system.
- If necessary, the presentation layer translates between **multiple data formats** by using a common format.
- Provides encryption and compression of data.
- Examples :- JPEG, MPEG, ASCII, EBCDIC, HTML.

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Layer 5: Session

- The session layer defines **how to start**, **control** and **end conversations** (called sessions) between applications.
- This includes the control and management of **multiple bi-directional messages** using **dialogue control.**
  - **keeping track of whose turn it is to transmit**
- It also synchronizes dialogue between two hosts' presentation layers and manages their data exchange.
  - preventing two parties from attempting the **same critical operation at the same time**
- The session layer offers provisions for efficient data transfer.
  - Check pointing long transmissions to allow them to continue from where they were after a crash

Examples :- SQL, ASP(AppleTalk Session Protocol).

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Layer 4: Transport

- It accepts data from above, **splits it up into smaller units if need be**, passes them to the network layer, and ensure that the pieces all arrive correctly at the other end
- The transport layer **segments data from** the sending host's system and reassembles the data into a data stream on the receiving host's system.
- **End-to-end error free delivery of entire message**
- Services include:
  - Service port addressing
    .Port number
  - Segmentation /reassembly
  - Connection control
    .Connectionless or connection oriented
  - Flow and error control

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Layer 3: Network

- Defines *end-to-end delivery of packets*.
- *Defines logical addressing* so that any endpoint can be identified.
- Defines ***how routing works*** and ***how routes are learned*** so that the packets can be delivered.
- The network layer also ***defines how to fragment a packet*** into smaller packets to accommodate different media.
- Examples :- IP, IPX, AppleTalk.

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Layer 2: Data Link

- The data link layer provides **access to the networking media** and **physical transmission across the media** and this enables the data to **locate its intended destination on a network.**
- The data link layer provides reliable transit of data across a physical link by using the **Media Access Control (MAC) addresses.**
- The data link layer uses **the MAC address** to define a hardware or data link address in order for multiple stations to share the same medium and still uniquely identify each other.
- Concerned with network topology, network access, error notification, ordered delivery of frames, and flow control.
- Examples :- Ethernet, Frame Relay, FDDI.

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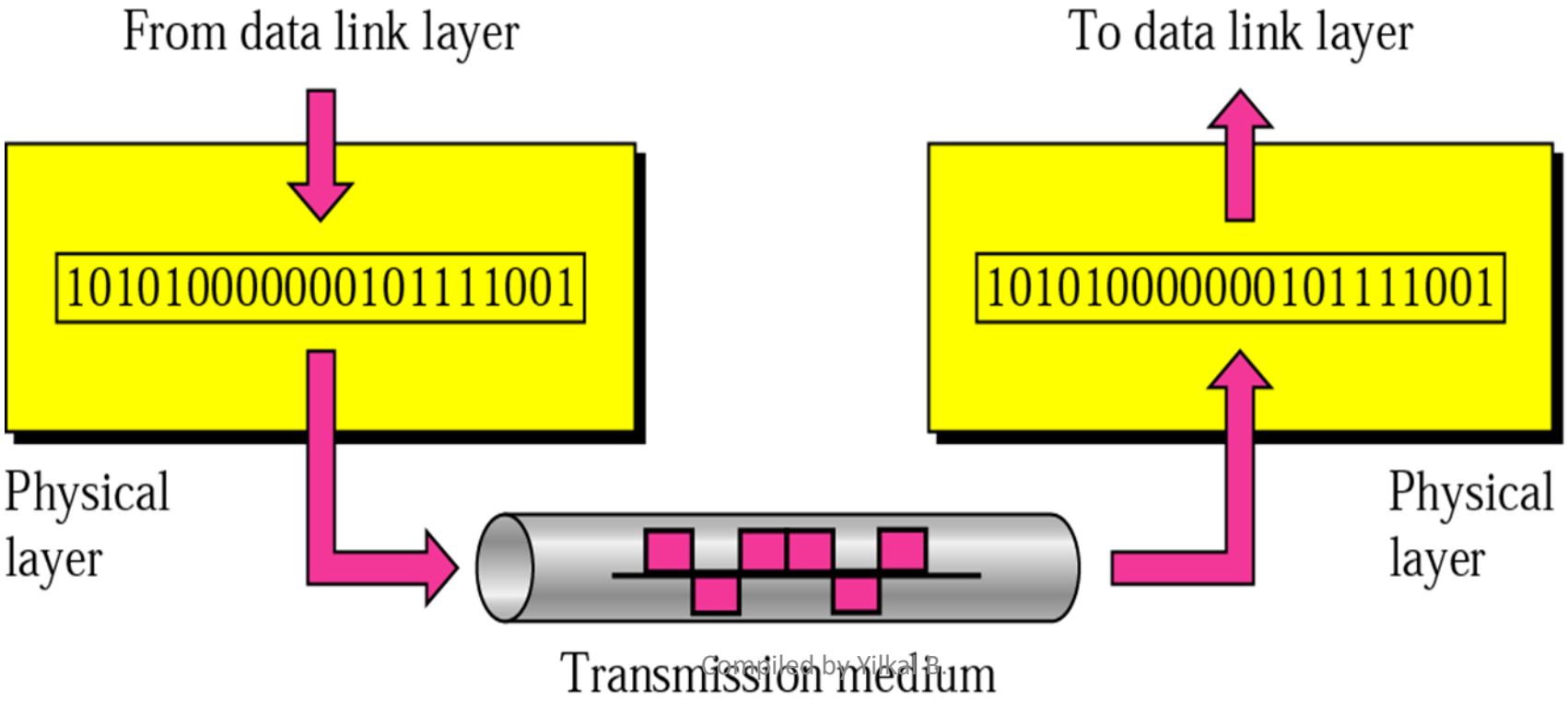
Layer 1: Physical

- The layer **that actually interacts with the transmission media**
- the physical part of the network that connects network components together
- involved in physically carrying information from one node in the network to the next
- The physical layer deals with the physical characteristics of the **transmission medium.**
- **It defines**
  - **mechanical:** the size and shape of the network connector, how many pins does the network connector has and what each pin is used for
  - **electrical:** how many volts represent a 1 and how many a 0
  - **timing**: how many nanoseconds a bit lasts
  - Whether communication is o**ne way** or in **both directions simultaneously**

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- *Such characteristics as voltage levels, timing of voltage changes, physical data rates, maximum transmission distances, physical connectors, and other similar attributes are **defined by physical layer specifications**.*

  Examples :- EIA/TIA-232, RJ45, BNC



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Shortcomings of OSI

- Bad Technology
- Session and presentation almost empty
- addressing, flow control, and error control, reappear again and again in each layer
- Bad implementation
- Bad timing
- Bad politics

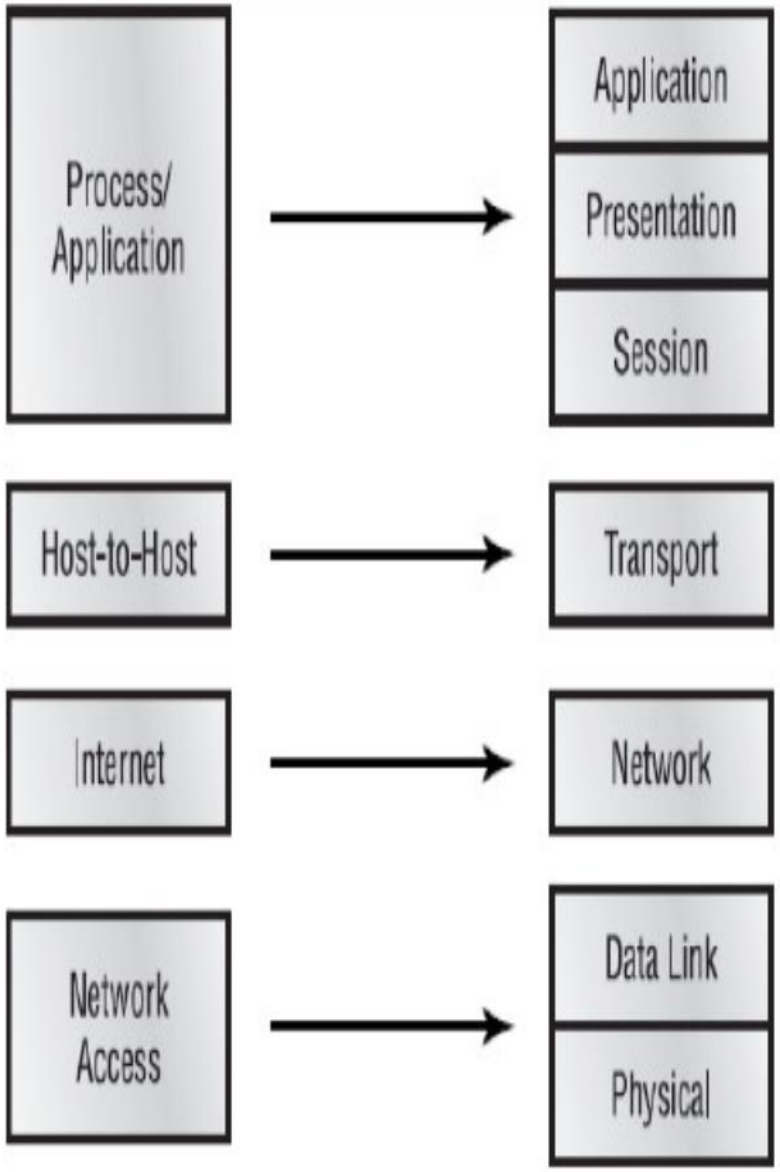
Cntd…

Summary of OSI model

- There was **no standard for networks** in the early days and as a result it was difficult for networks to communicate with each other.
- The International Organization for Standardization (ISO) recognized this. and researched various network schemes, and in 1970 introduced the Open Systems Interconnection (OSI) reference model.
- **The OSI reference model** has standards which ensure vendors (sellers) greater compatibility and interoperability between various types of network technologies.
- The OSI reference model organizes network functions into **seven numbered layers**.
- Each layer provides a service to the layer above it in the protocol specification and communicates with the same layer's software or hardware on other computers.
- Layers 1-4 are concerned with the **flow of data from end -to- end** through the network and Layers 5-7 are concerned **with services to the applications**.

TCP/IP Reference Model

- used in the grandparent of all wide area computer networks, the ARPANET
- Developed by **Department of Defense** (DoD) to ensure and preserve data integrity as well as maintain communication in the even of catastrophic war
- Condensed version of OSI model – contains four layers instead of seven



Functions of layer

Goals

- **The Process/Application :** protocols for **node-to-node application** communication and also co ntrols user-interface specifications.
- **The Host-to-Host**
  - parallels the functions of the OSI's Transport layer, defining protocols for setting up the le vel of transmission service for applications.
- **The Internet layer**
  - corresponds to the OSI's Network layer, designating the protocols relating to the logical tr ansmission of packets over the entire network.
- **Network Access layer**
  - The equivalent of the **Data Link and Physical layers** of the OSI model
  - Oversees hardware addressing and defines protocols for the physical transmission of data.

Created by WPS Office

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Shortcomings of TCP/IP

- the model **does not clearly distinguish the concepts of service, interface, and p rotocol**
- the TCP/IP model **is not at all general** and **is poorly suited to describing** any protocol stack other than TCP/IP
- the host-to-network layer is not really a layer at all
- **does not** distinguish (or even mention) the physical and data link layers

OSI vs. TCP/IP

In OSI:

- *Network –Connectionless/ Connection oriented*
- *Transport –Only Connection oriented*

In TCP/IP:

- *Transport - Connectionless/ Connection oriented*
- *Very useful for simple request reply*
- *Network – Only Connectionless*

OSI:

- *Difficult to Implement*

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OSI: Service, Interfaces and protocols

- Layers Interface: How layer above it access it, what parameter and results to expect
- Peer protocols: Used in a layer are the layer's business
- Layer is equivalent to an Object
  - Set of methods

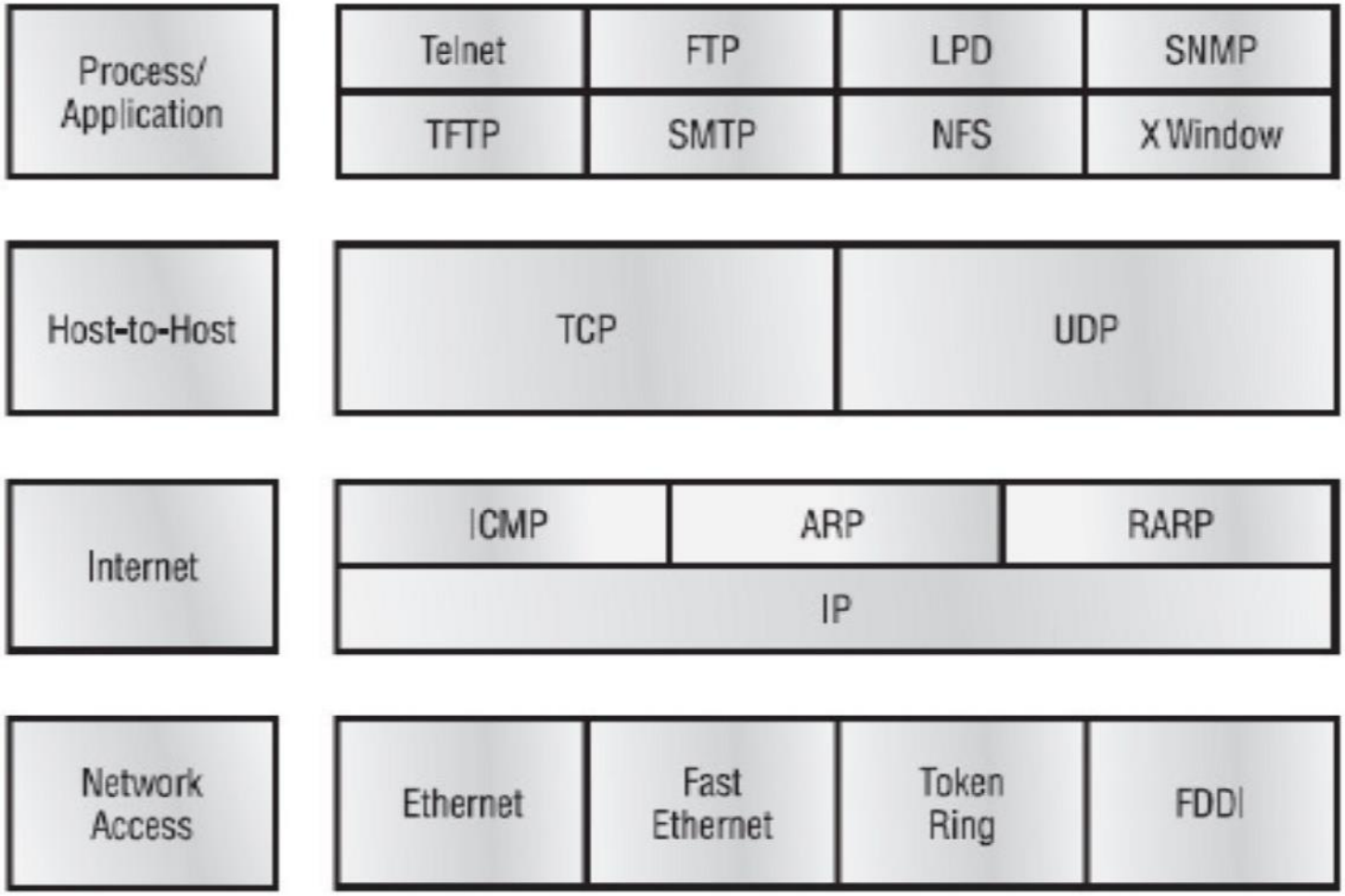
TCP/ IP–no distinction between protocol and service

IP :- Send IP packet and Receive IP packet

Host - to - network (TCP/IP)

- Not really a layer. Interface between network and data link layer
- No distinction between physical and data link layer
- **Adhoc application layer protocols**
  - ELNET: Virtual terminal designed for a character terminal
  - no more than a UI

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Process/ Application
Telnet
FTP
LPD
SNMP
TFTP
SMTP
NFS
X Window
Host-to-Host
TCP
UDP
Internet
ICMP
ARP
RARP
IP
Network Access
Ethernet
Fast Ethernet
Token Ring
FDDI

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Application/Process layer

- ***Telnet*** *- allows a user on a remote client machine, called the Telnet client, to access the resources of another machine, the Telnet server.*
  - *Telnet makes client machine appear as though it were a terminal directly attached to the server.*
- ***File Transfer Protocol (FTP)*** *- is the protocol that actually lets us transfer files, and it can accomplish this between any two machines using it.*
  - *Usually users are subjected to authentication*
- ***Network File System (NFS)*** *- a protocol specializing in file sharing allowing two different types of file systems to interoperate.*

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- ***Simple Mail Transfer Protocol (SMTP)*** *- uses a spooled, or queued, method of mail delivery.*
  - *POP3 is used to receive mail.*
- ***Simple Network Management Protocol (SNMP)*** *- collects and manipulates valuable network information.*
  - *This protocol stands as a watchdog over the network, quickly notifying managers of any sudden turn of events.*
- ***Domain Name Service (DNS)*** *– resolves hostnames—specifically, Internet names, such as www.amu.edu.et to the IP address 10.6.10.3*
- ***Dynamic Host Configuration Protocol (DHCP)*** *- gives IP addresses to hosts.*
  - *It allows easier administration and works well in small-to-even-very large network environments.*

Host-to-Host layer

- **Transmission Control Protocol (TCP)** - takes large blocks of information from an application and breaks them into segments.
  - It numbers and sequences each segment so that the destination's TCP protocol can put the segments back into the order the application intended.
  - Uses three way handshaking
- **User Datagram Protocol (UDP)** - does not sequence the segments and does not care in which order the segments arrive at the destination. But after that, UDP sends the segments off and forgets about them. It doesn't follow through, check up on them, or even allow for an acknowledgment of safe arrival — complete abandonment.
  - TCP for reliability and UDP for faster transfers.

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- ***TCP and UDP*** *must use port numbers to communicate with the upper layers, because they're what keeps track of different conversations crossing the network simultaneously.*
- ***These port numbers*** *identify the source and destination application or process in the TCP segment.*
- *There are 2^16 = 65536 ports available.*
  - ***Well-known ports*** *- The port numbers range from 0 to 1023.*
  - ***Registered ports*** *- The port numbers range from 1024 to 49151.*
    - *Registered ports are used by applications or services that need to have consistent port assignments.*
  - ***Dynamic or private ports*** *- The port numbers range from 49152 to 65535. These ports are not assigned to any protocol or service in particular and can be used for any service or application.*

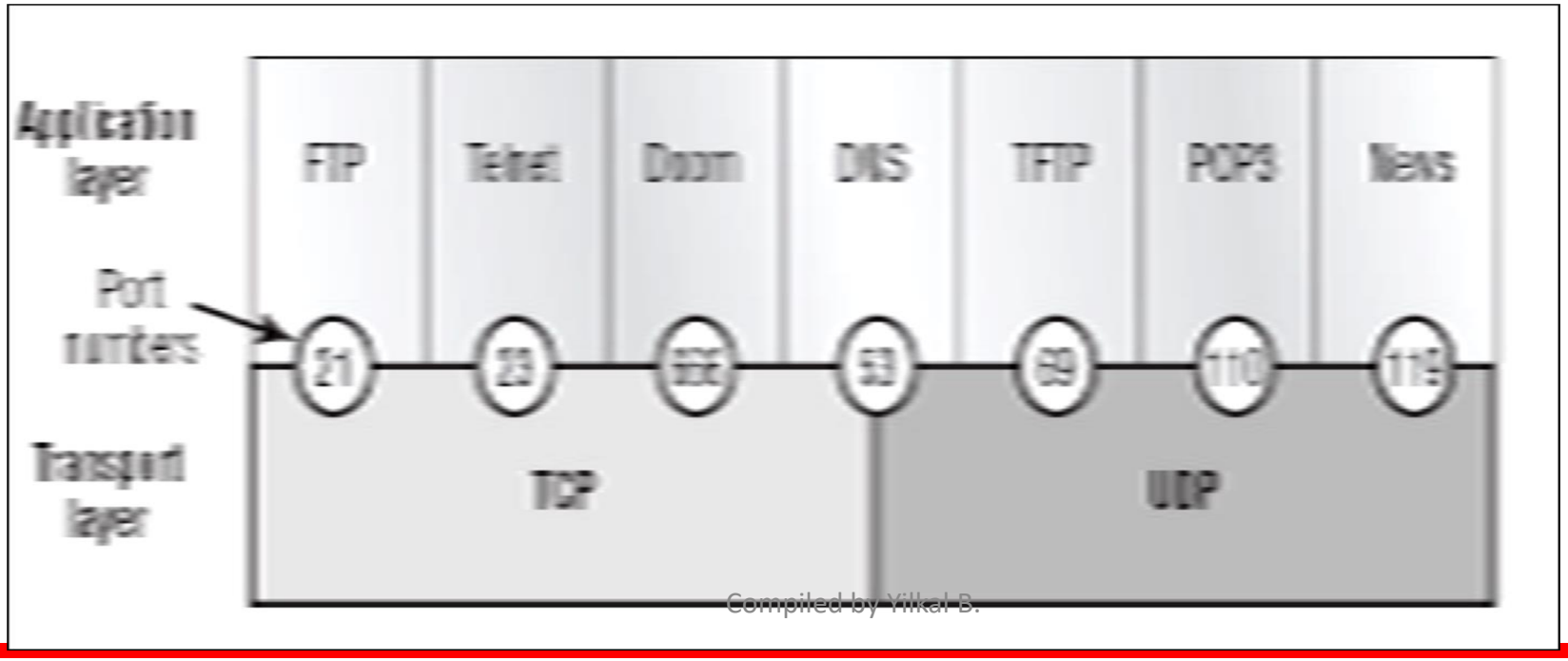
Cntd…

- *If a port is **closed/blocked,** you **cannot communicate** with the computer by the protocol using that port.*

  Eg. If port 25 is blocked you cannot send mail.

  Firewalls by default block all ports.

  You should know the port numbers of different protocols!!



Internet Layer

- ***Internet Protocol (IP)*** *essentially is the Internet layer.*
- *The other protocols found here merely exist to support it.*
- *It can do this because all the machines on the network have a software, or logical, address called an IP address.*
- ***Internet Control Message Protocol (ICMP)*** *works at the Network layer and is used by IP for many different services.*
  - *ICMP is a management protocol and messaging service provider for IP.*
  - *The following are some common events and messages that ICMP relates to:*
    - ***Destination Unreachable*** *If a router can't send an IP datagram any further, it uses ICMP to send a message back to the sender, advising it of the situation.*
    - ***Buffer Full*** *If a router's memory buffer for receiving incoming datagrams is full, it will use ICMP to send out this message until the congestion abates.*

Cntd…

- *Hops Each IP datagram is allotted a certain number of routers,* ***called hops,*** *to pass through. If it reaches its limit of hops before arriving at its destination, the last router to receive that datagram deletes it.*
- ***The executioner router then uses ICMP*** *to send a message, informing the sending machine of the end of its datagram.*
- ***Ping*** *(Packet Internet Groper) uses ICMP echo messages to check the physical and logical connectivity of machines on a network.*
- ***Trace route*** *Using ICMP timeouts, Traceroute is used to discover the path a packet takes as it traverses an internetwork.*

➢ ***Address Resolution Protocol (ARP)*** *finds the hardware address of a host from a known IP address.*

- *ARP interrogates the local network by sending out a broadcast asking the machine with the specified IP address to reply with its hardware address.*

Cntd…

Reverse Address Resolution Protocol (RARP)

- *discovers the identity of the IP address for diskless machines by sending out a packet that includes its MAC address and a request for the IP address assigned to that MAC address.*
- *A designated machine, called a RARP server, responds with the answer, and the identity crisis is over.*



Compiled by Vikal Poudel

Network Access

- *Wi-Fi(wireless fidelity)*
- *ATM*
- *PPP*
- *Ethernet*
- *Fast Ethernet*
- *FDDI*
- *Token ring*

- ***Network Access technology detail explanation is reading assignment***

End of chapter Two

Any question??